American Computer Science League

2025-2026 ● Contest 2: Short Problems Solutions ● Junior Division

1. Prefix/Infix/Postfix Notation

$8 - 3 * 2 + 2^3 - 8 / 2$
= 8 - (* 3 2) + (^2 3) − (/ 8 2)
= (- 8 (* 3 2)) + (^2 3) − (/ 8 2)
= (+ (− 8 (* 3 2)) (^ 2 3)) − (/ 8 2)
= − (+ (− 8 (* 3 2)) (^ 2 3)) (/ 8 2)
= − + − 8 * 3 2 ^ 2 3 / 8 2

C. − + − 8 * 3 2 ^ 2 3 / 8 2

2. Prefix / Infix / Postfix

Evaluate the following postfix expression if a = 3, b = -2, and c = 5?

a c + a b − * a b − c + / 2 ^
= ((a + c) * (a - b) / (a - b + c)) ^ 2
= ((3 + 5) * (3 - (-2)) / (3 - (-2) + 5)) ^ 2
= (8 * 5 / 10) ^ 2
= (40 / 10) ^ 2
= 16

D. 16

3. Bit-String Flicking

NOT (10110 OR 11100 AND NOT 01101)
= NOT (10110 OR 11100 AND 10010)
= NOT (10110 OR 10000)
= NOT 10110
= 01001

E. 01001

4. Bit-String Flicking

(RCIRC-3 ((LCIRC-2 01101) AND (RSHIFT-2 11010)))
= (RCIRC-3 (10101 AND 00110))
= (RCIRC-3 00100)
= 10000

B. 10000

5. What Does This Program Do? (Loops)

```
count = 0
for x = 1 to 9
    num = x*100+x*10+x
    sum = 3*x
    if num/sum==37 then
        count = count+1
    end if
next x
```

This program calculates the values of 111/3, 222/6, 333/9, 444/12, 555/15, 666/18, 777/21, 888/24, and 999/27. Amazingly, every value equals 37. Not all need to be done since there is clearly a pattern and the choices are 0, 1, 3, 5, 9.

E. 9

6. What Does This Program Do? (Loops)

```
num = 1089
for x = 3 to 11
    count = 0
    while num % x == 0
        count = count + 1
        num = num / x
    end while
    if count != 0 then
        output x, count
    end if
next x
```

This program finds the prime factorization of the number 1089:
1089 / 3 = 363
363 / 3 = 121
121 is not divisible by 4, 5, 6, 7, 8, 9, or 10.
121 / 11 = 11
11 / 11 = 1
so 1089 = 3^2 * 11^2.

B.
```
3 2
11 2
```